Modeling Repeats and Diploidy in Low Coverage, Reference-Free, Population Genomics

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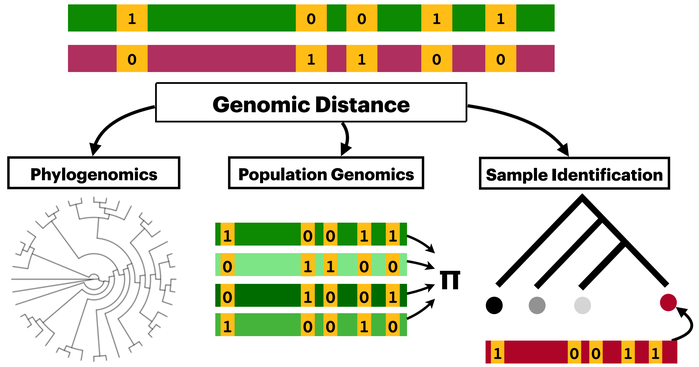
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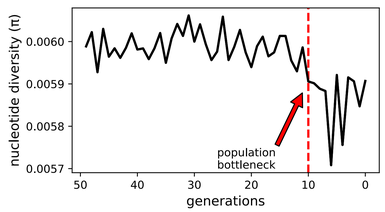
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Background: Genomic Distance and Pop Gen

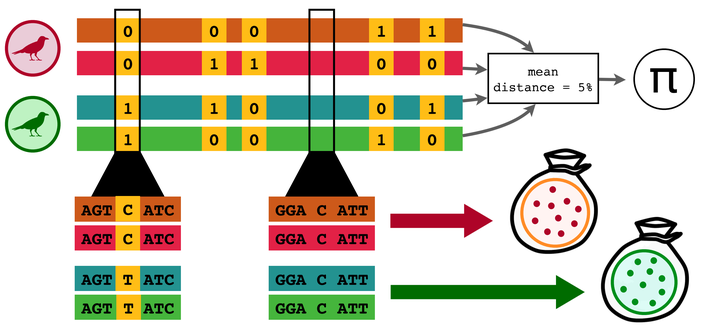
- **Genomic distance** is a simple metric with a variety of applications.



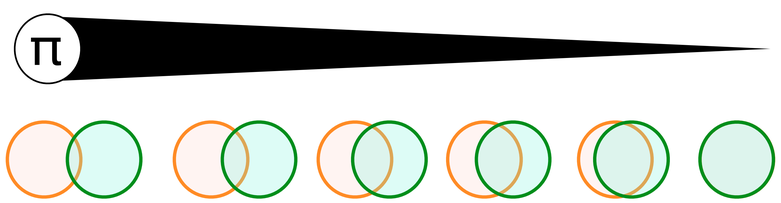
- World Wildlife Foundation: **73% decline in biodiversity** (1970-2022)
- **Nucleotide diversity** is one way to monitor biodiversity, but many samples across time must be collected **(expensive)**.



- We can also measure π through 'bags of *unique* *k*-mers' (sets of *k*-mers).



- The lower π is, the more similar *k*-mer sets will be (Jaccard Index).

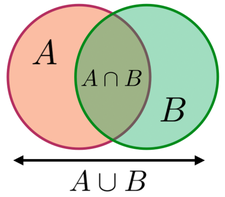


Genome Skimming for Accessible Pop Gen

- By using low-coverage sequencing, we can perform accessible temporal whole-genome sampling of populations.
- **Mash** (*Ondov et al. 2016*), **Skmer** (*Sarmashghi et al. 2019*), and **ReSkmer** (*Charvel and Thomas et al., 2026*) all use the *k*-mer similarity in genome skims to estimate distance.

$$J = \frac{|A \cap B|}{|A \cup B|}$$

Jaccard Similarity



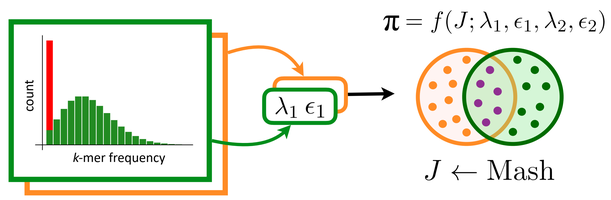
$$d = 1 - \left(\frac{2J}{J+1}\right)^{1/k}$$

Genetic Distance

- *Genome Skimming* is a technique that lets you obtain a sparse random sample of short nucleotide sequences spanning the genome.

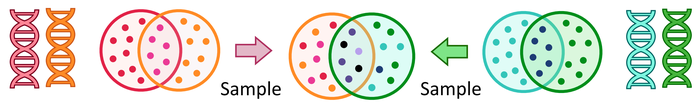


- These methods use *k*-mer counts in skims to estimate coverage (λ) and error (ϵ) and incorporate them into final π estimates.

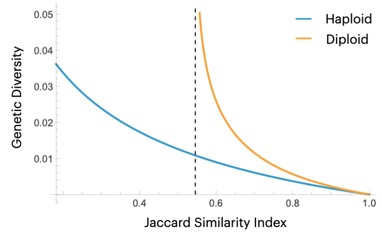


Diploid Genome Skims and Jaccard Similarity

- Existing methods don't incorporate heterozygosity into *k*-mer similarity.

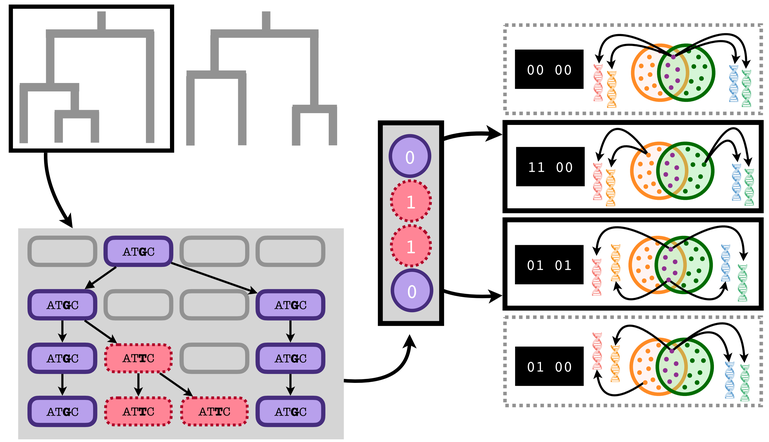


- The Jaccard Index translates to very different genetic diversity (π) values depending on whether you compare diploid or haploid samples.

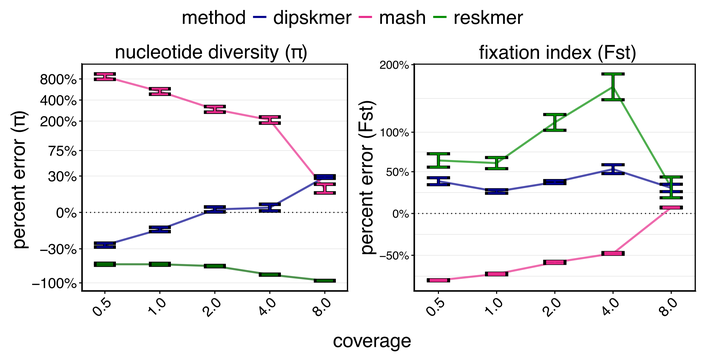


Diploid Model of Genomic Distance

1. Probability model for *k*-mer tree topologies across 4 chromosomes.
2. Mutation probability model along a k-mer genealogy.
3. Independent sorting probability of k-mers into samples.



Results



- DipSkmer improves upon other genome skimming methods in the estimation of different population genetic metrics, such as nucleotide diversity (π) and divergence between populations (F_{st}).

Published in PLOS Computational Biology

- **Summary:** DipSkmer can use affordable low-coverage sequencing to estimate *genetic diversity* and infer *population structure* with the same accuracy as high-coverage methods.
- Scan Here for Paper ⇒

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